CHAPTER 4

Network Layer

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4.1 Introduction

- **Network Layer Services:** Routing, Forwarding

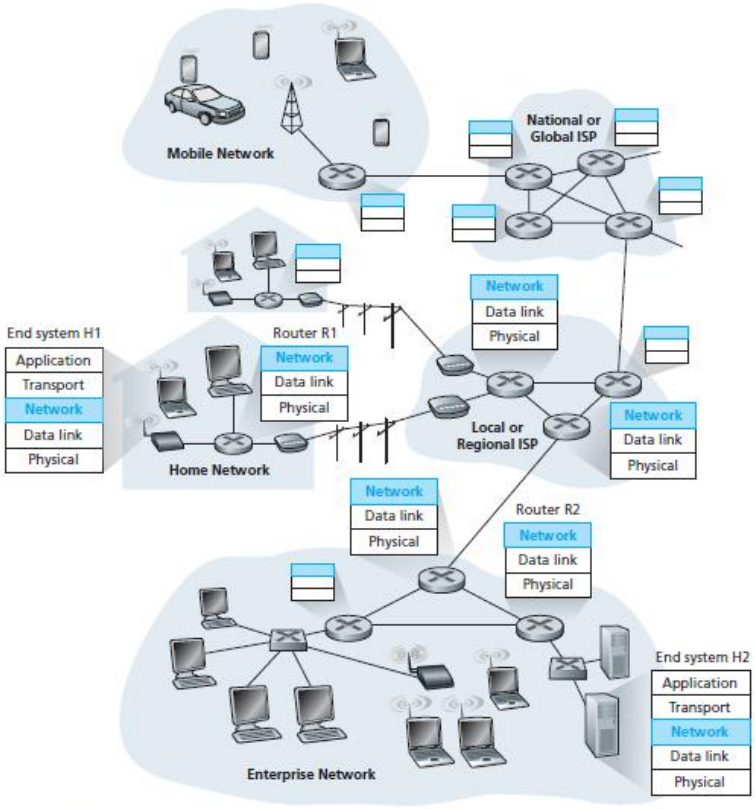


Figure 4.1 ♦ The network layer

- **Routing algorithms and Forwarding table**

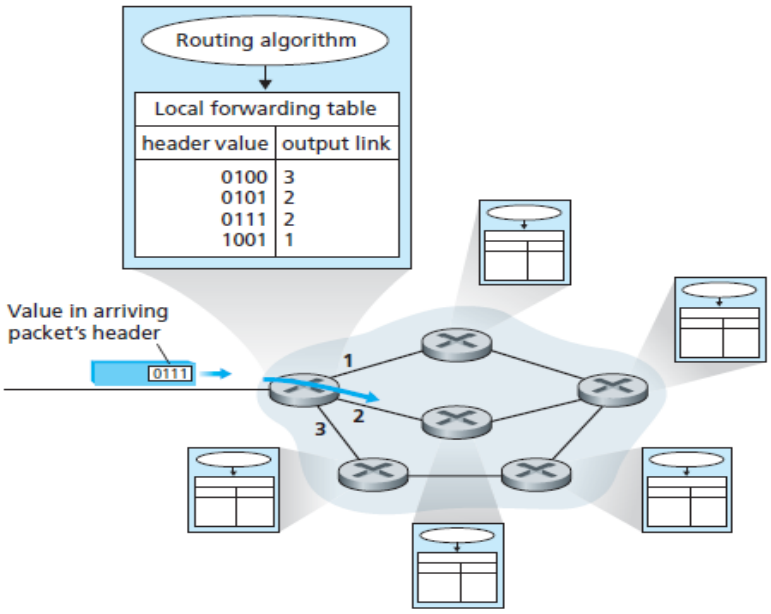
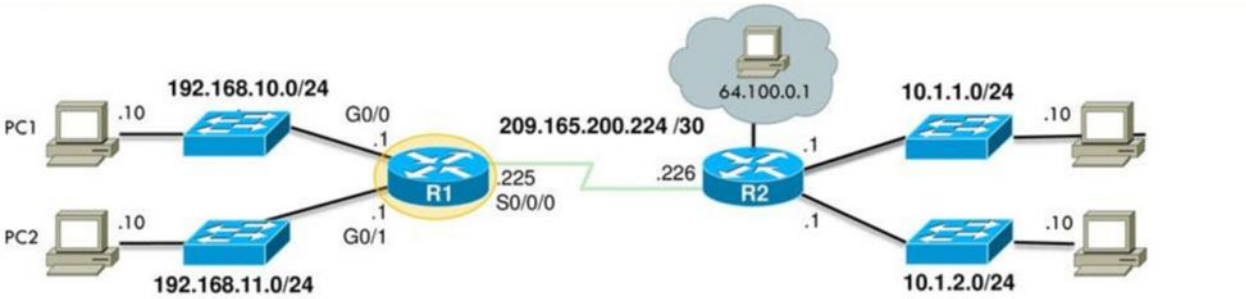


Figure 4.2 ♦ Routing algorithms determine values in forwarding tables

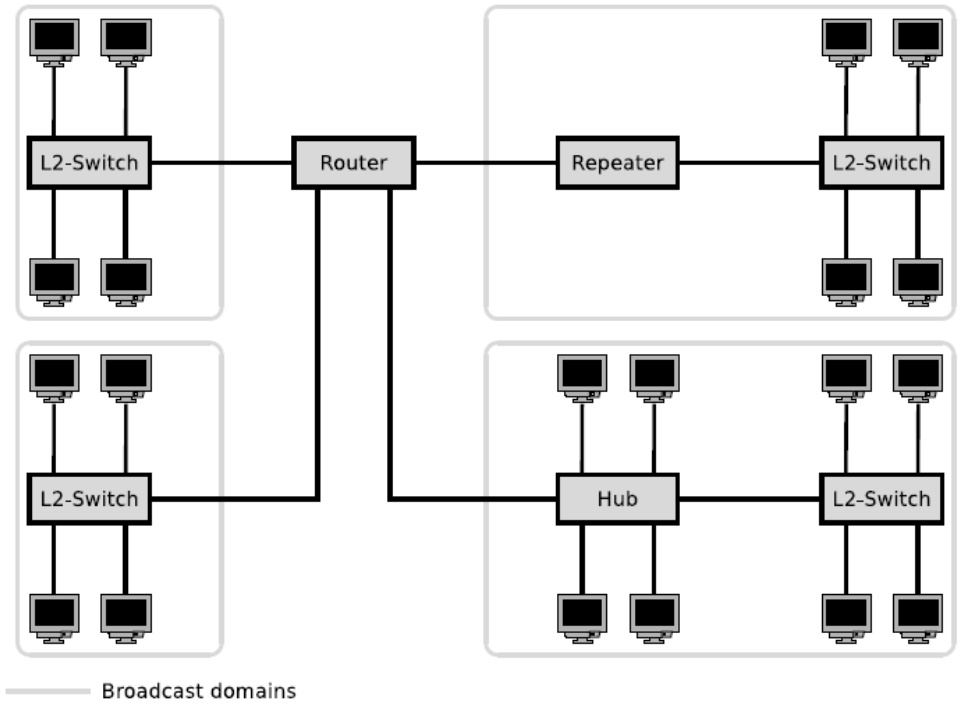



```
A          B                                                C
C   192.168.10.0/24 is directly connected, GigabitEthernet0/0
L   192.168.10.1/32 is directly connected, GigabitEthernet0/0
```

A	Identifies how the network was learned by the router.
B	Identifies the destination network and how it is connected.
C	Identifies the interface on the router connected to the destination network.

❑ Exercise 1: Broadcast Domain

1. Sketch in the diagram of the network topology all **broadcast domains**.
2. What is the **required number of subnets** for this network topology?



4.2 Virtual Circuit and Datagram Networks

- **Two classes of Computer Networks:** Virtual Circuit & Datagram Networks

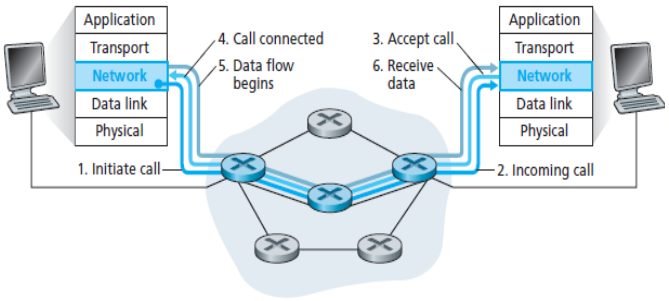


Figure 4.4 ♦ Virtual-circuit setup

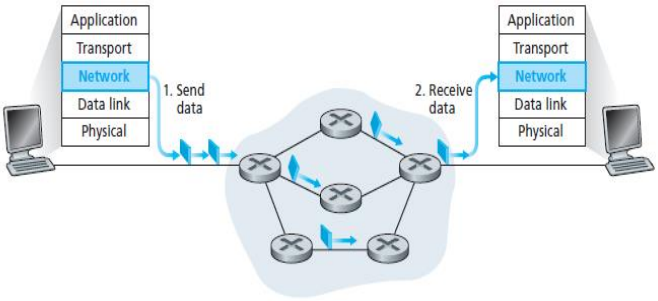


Figure 4.5 ♦ Datagram network

- **A simple Virtual Circuit example** (ATM, Frame Relay)

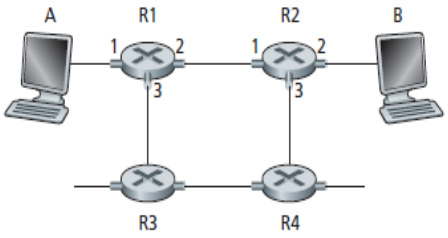


Figure 4.3 ♦ A simple virtual circuit network

Incoming Interface	Incoming VC #	Outgoing Interface	Outgoing VC #
1	12	2	22
2	63	1	18
3	7	2	17
1	97	3	87
...	...	...	...

❑ Exercise 1:

Using the example network given in Figure 3.44, give the **virtual circuit tables** for all the switches after each of the following connections is established. Assume that the sequence of connections is **cumulative**; that is, the first connection is still up when the second connection is established, and so on. Also, assume that the Virtual Circuit ID (VCI) assignment always picks the **lowest unused VCI** on each link, starting with 0, and that a VCI is consumed for **both directions** of a virtual circuit.

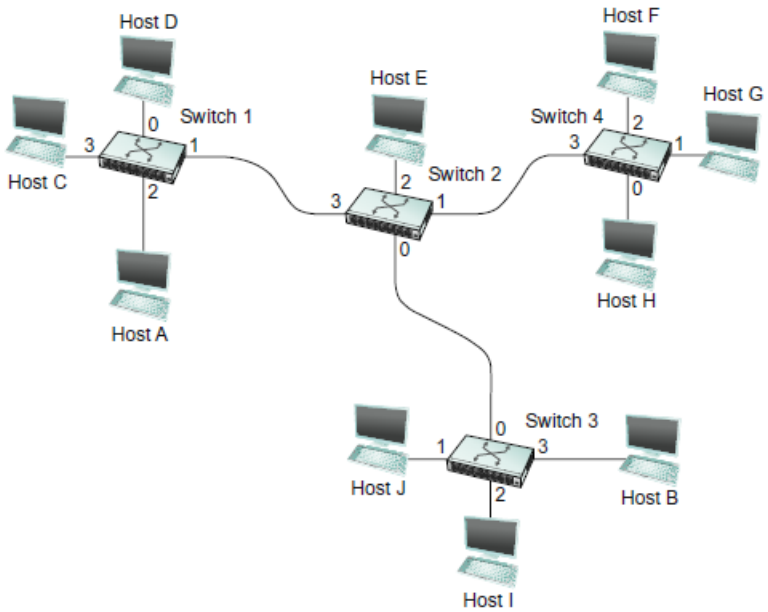


FIGURE 3.44 Example network for Exercises 1 and 2.

a) Host D connects to host H.
b) Host B connects to host G.
c) Host F connects to host A.
d) Host H connects to host C.
e) Host I connects to host E.
f) Host H connects to host J.

❖ **Datagram Network example** (Internet)**, Longest prefix matching**

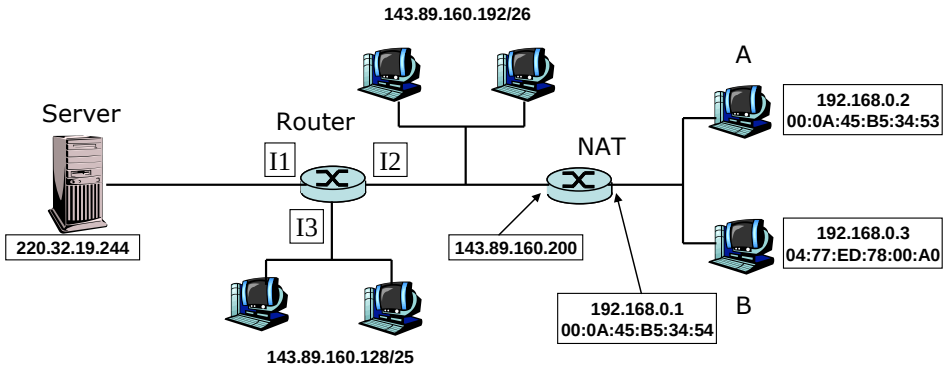
Destination Address Range	Link Interface
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

Prefix Match	Link Interface
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
otherwise	3

✓ Which interface: **11001000 00010111 00010110 10100001**
✓ Which interface: **11001000 00010111 00011000 10101010** (Use **longest prefix matching**)

❑ Exercise 2:

Consider a network setting in the following figure. Node *A* and node *B* are in a LAN behind the NAT.



Answer the following questions:

a) Show the **routing table** at router named as **"Router"** with entries for the following fields:

Network Address	Subnet Mask	Next Hop (I1 / I2 / I3)

b) Describe with **which interface** the packets of the following destinations will be delivered:
i. 200.32.19.201 ii. 143.89.160.197 iii. 143.89.160.145

Knowledge Test # 1

Q.1 Compared to **two upper layers** (Application and Transport layer) what is the fundamental difference between **network layer** and **those two layers** in respect of **figure 4.1** in the textbook?

Q.2 What are the **two main functions** of **network layer**? Illustrate with **an example** (**Hint:** use **figure 4.2** in textbook).

Q.3 Name the **three private IP address spaces**.

Q.4 What is the difference between **virtual circuit** and **datagram networks**? Illustrate with proper diagrams (**Hint:** Use **figure 4.3, 4.4** and **4.5** in the textbook)

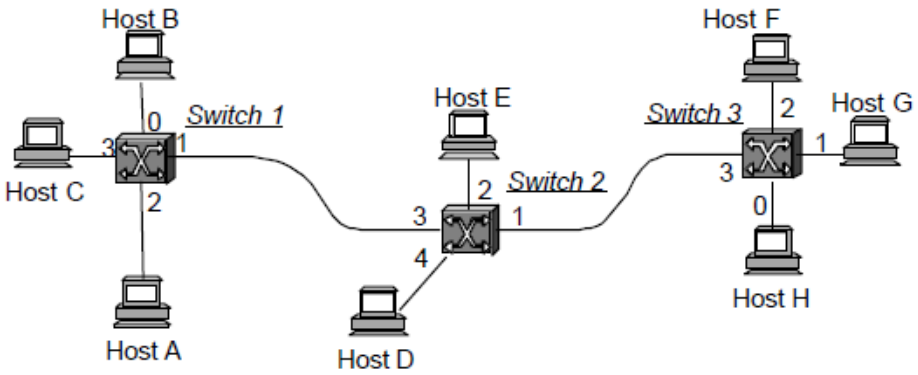
Q.5 Consider the **Virtual Circuit network** described in the Figure; the **current content** of the Virtual Circuits tables is given in the **switching tables**. Identify the connections already set up by listing their **end points**.

Switch 1			
In		Out	
Port	VCI	Port	VCI
0	0	1	1
2	0	1	2
3	0	1	0
3	1	1	3

Switch 2			
In		Out	
Port	VCI	Port	VCI
3	0	2	0
3	1	1	0
3	2	1	1
3	3	1	2

Switch 3			
In		Out	
Port	VCI	Port	VCI
3	2	0	0
3	0	1	0
3	1	2	0

Table 2: Switching Tables



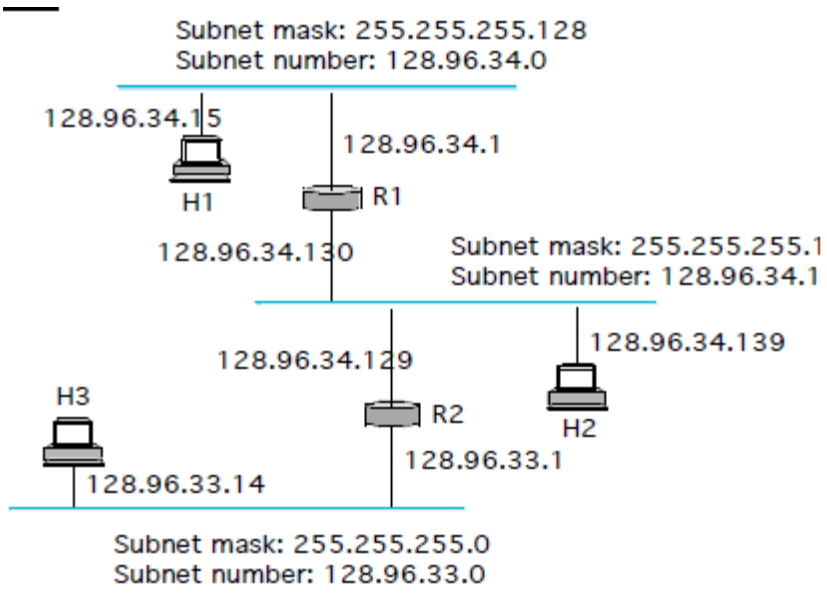
Q.6 What is a **forwarding/routing table**? Why we need **longest prefix matching** in datagram networks?

Q.7 Suppose a router has built up the **routing table** shown in the following table. The router can deliver packets directly over interfaces 0 and 1, or it can forward packets to routers R2, R3, or R4. Assume, the router does the longest prefix match.

SubnetNumber	SubnetMask	NextHop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.252.0	R3
⟨default⟩		R4

a. Describe what the router does with a packet addressed to each of the following destinations:
i.128.96.171.92, ii.128.96.167.151 iii.128.96.169.192 iv. 128.96.165.121

Q.8 Assume subnetting is used. Write routing table at R2 for network in picture, assuming following entry format: **Dest | NextHop | Subnet Mask**



4.4 The Internet Protocol (IP): Forwarding & Addressing in Internet

❖ IP Datagram format

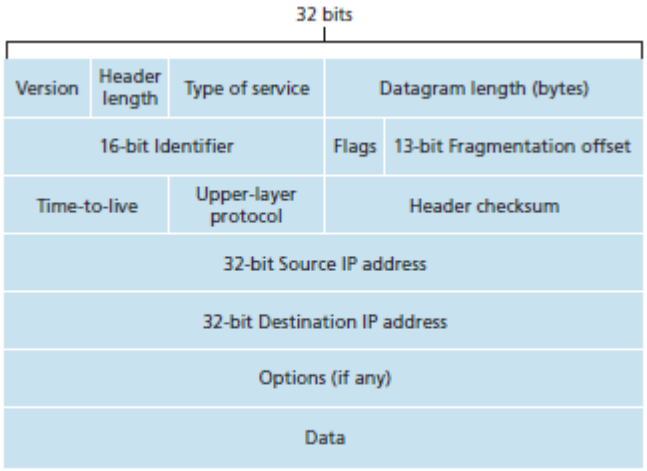


Figure 4.13 ♦ IPv4 datagram format

❖ IP Datagram fragmentation

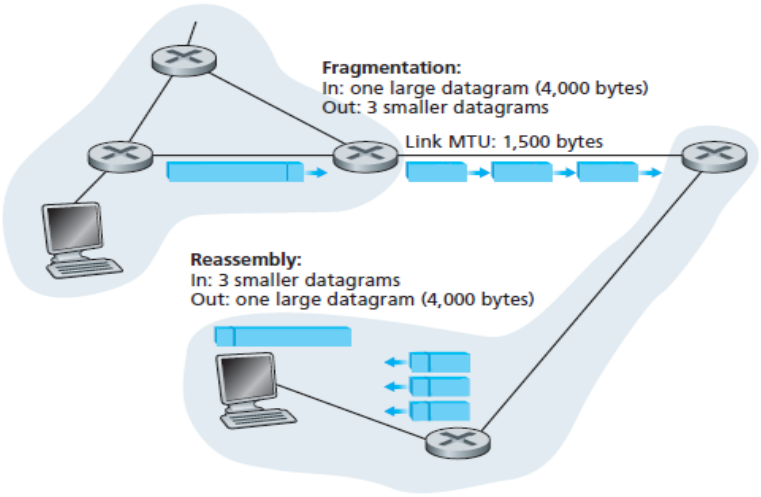


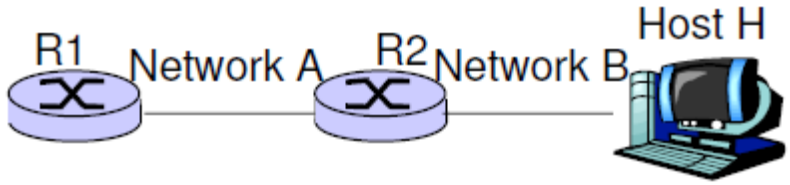
Figure 4.14 ♦ IP fragmentation and reassembly

Fragment	Bytes	ID	Offset	Flag
1st fragment	1,480 bytes in the data field of the IP datagram	identification = 777	offset = 0 (meaning the data should be inserted beginning at byte 0)	flag = 1 (meaning there is more)
2nd fragment	1,480 bytes of data	identification = 777	offset = 185 (meaning the data should be inserted beginning at byte 1,480. Note that 185 · 8 = 1,480)	flag = 1 (meaning there is more)
3rd fragment	1,020 bytes (= 3,980–1,480–1,480) of data	identification = 777	offset = 370 (meaning the data should be inserted beginning at byte 2,960. Note that 370 · 8 = 2,960)	flag = 0 (meaning this is the last fragment)

Table 4.2 ♦ IP fragments

❑ Exercise 3:

Consider the following scenario:



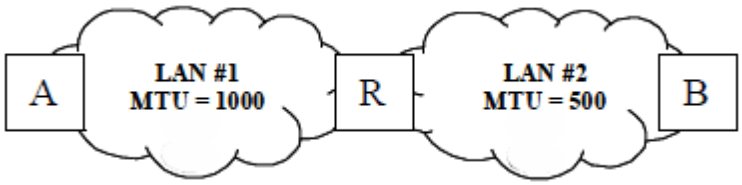
Assume an **MTU OF 1500 Bytes** for network A and **532 bytes for network B** respectively. Assume R1 receives an IP packet [packet P] directed at H and containing a **2000 bytes TCP segment** [TCP header + data]. Assume 20 bytes for IP header.

i. Describe the **M bit [More Fragment]** and the **Offset field** for IP fragments of packet P sent from R1 on network A
ii. Answer the same question packets sent from R2 over network B .

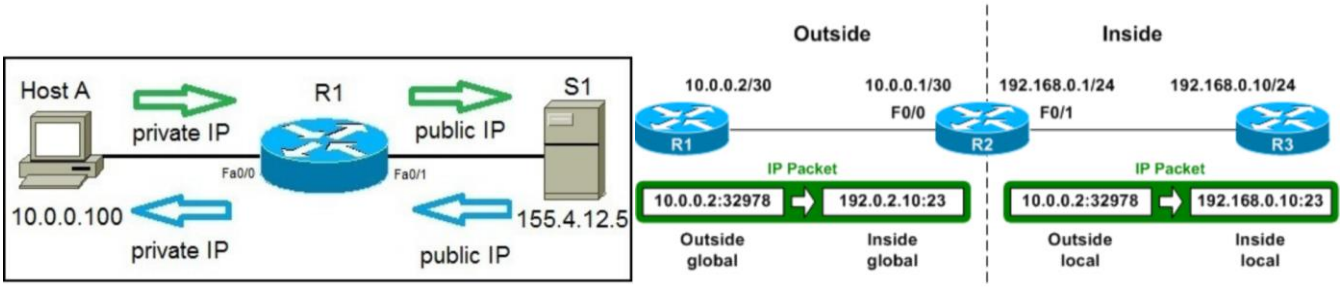
❑ Exercise 4:

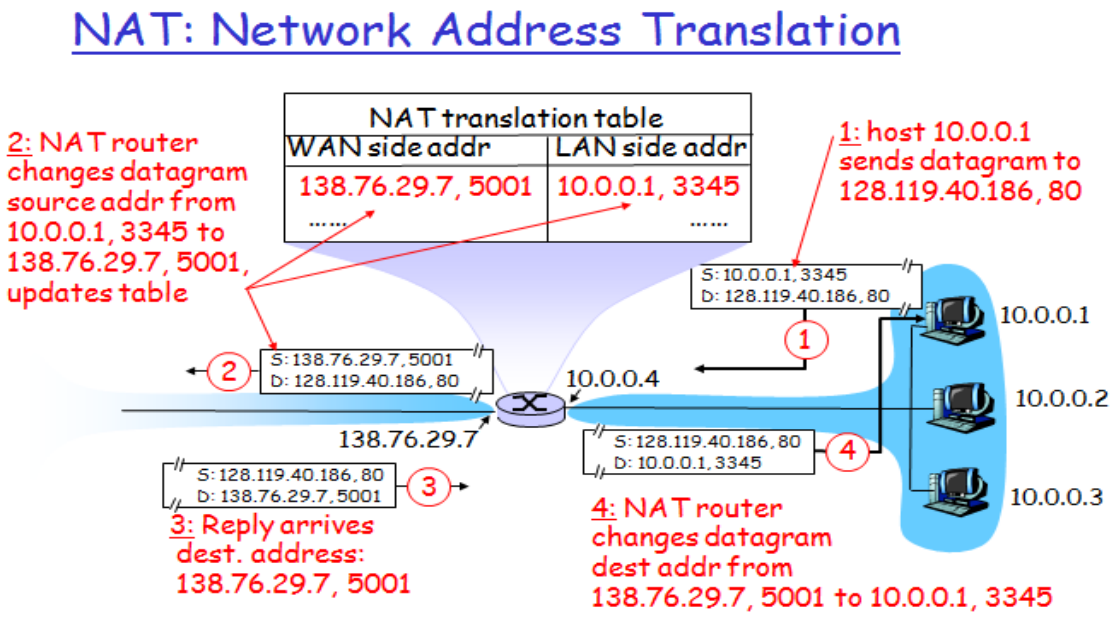
Host A is on LAN #1 and Host B is on LAN #2. The two LANs are interconnected via a router. The MTU of LAN #1 and LAN #2 are 1000 and 500 bytes respectively. Suppose an application on Host A executes 1000 writes to an application running on host B; each write results in one MTU-sized IPv4 packet on LAN #1. Each packet consists of only an IP header and data and the IP header is 20 bytes (no options).

Question: How many packets from Host A to Host B traverse LAN #2 ?



❖ Network Address Translation (NAT)





❑ Exercise 5:

Consider the scenario in **Figure 4.22** on **page 360** of the text. Suppose that **host 10.0.0.2** initiates a connection, using **source port 5500** to a **web server listening at port 80 at 128.119.40.186**.

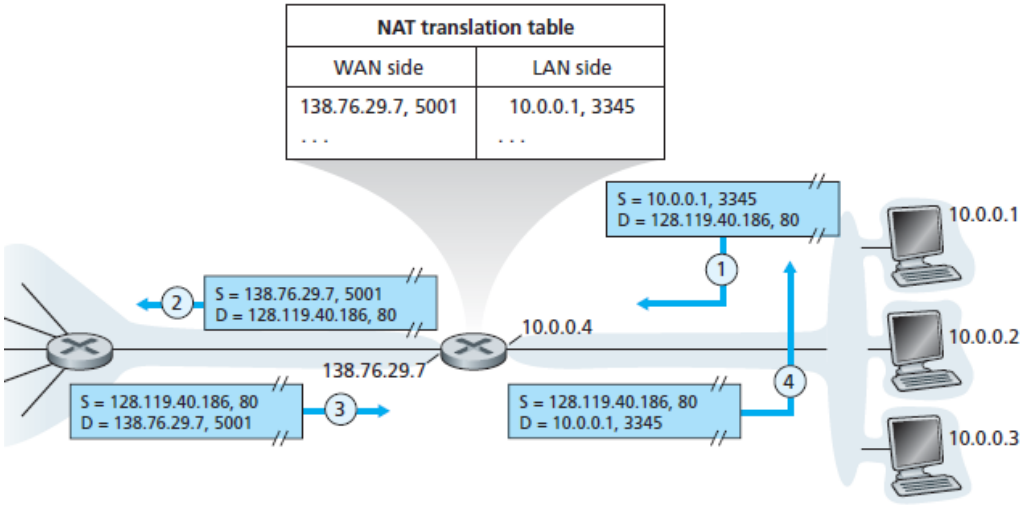


Figure 4.22 ♦ Network address translation

a) **Complete the NAT translation table** for this TCP connection.

b) What are the **source and destination IP addresses** and **port numbers** on the IP datagram arriving to the WAN side of the router with interface address **138.76.29.7**?

4.4.2 IPv4 Addressing (Already Covered)

Knowledge Test # 2

Q.1 Why the following fields are used in an **IP datagram**: Source & Destination IP address, ID, Flags, Fragmentation offset, time-to-live (TTL) and checksum.

Q.2 Why an **IP datagram** needs to be fragmented? How the fragments can be reassembled to produce the original datagram using **ID, Flags** and **Offset** fields? (**Hint:** Use **figure 4.14** in the textbook).

Q.3 Suppose a router receives a datagram of **5000 bytes**. But the outgoing ling has the capacity of frames of maximum **1500 bytes** (MTU = 1500 bytes) with header length **20 bytes**. Show how the datagram will be **fragmented.** (**Hint:** Show values of the fields: ID, fragFlag and Offset).

Q.4 IP Packets on a certain network can carry a **maximum of only 500 bytes** in the data portion. An application using TCP/IP on a node on this network generates a TCP segment with **1,000 bytes** in the data portion. **How many IP packets** are transmitted to carry this TCP segment, and what are their sizes (including the header)?

Q.5 Why **NAT** is used? What are the **advantages** and **disadvantages** of NAT? Illustrate **STATIC NAT** and **DYNAMIC NAT** with proper examples.

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4.5 Routing Algorithms

- **Classification of Routing algorithms:**
  - ✓ Link state & Distance vector (Global or Decentralized)
  - ✓ Static & Dynamic

- **Link State Algorithms** (Example: OSPF)

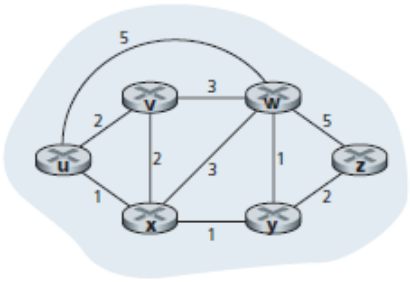


Figure 4.27 ♦ Abstract graph model of a computer network

step	N′	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5	uxyvwz					

Table 4.3 ♦ Running the link-state algorithm on the network in Figure 4.27

❖ **Distance vector Algorithms** (Example: RIP, EIGRP)

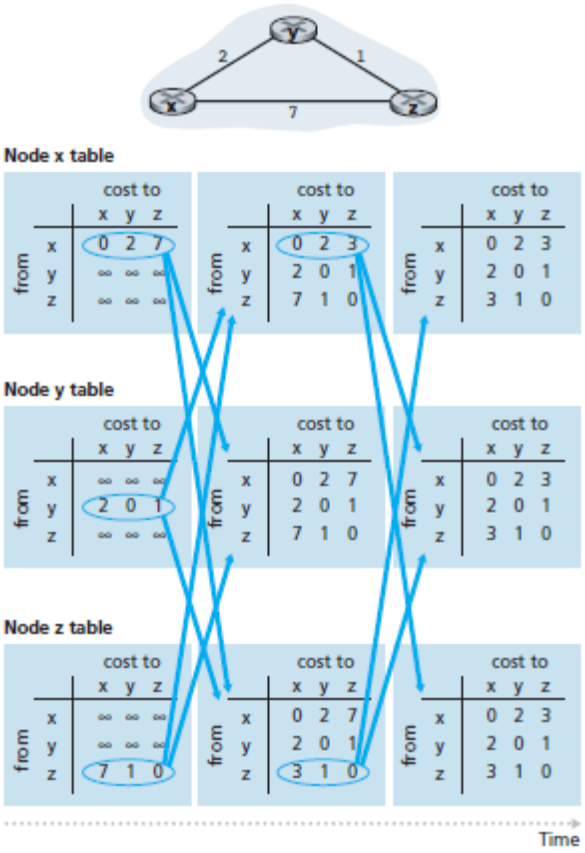


Figure 4.30 ♦ Distance-vector (DV) algorithm

4.6 Routing in the Internet

- **Autonomous Systems (AS): Intra-AS & Inter-AS Routing**

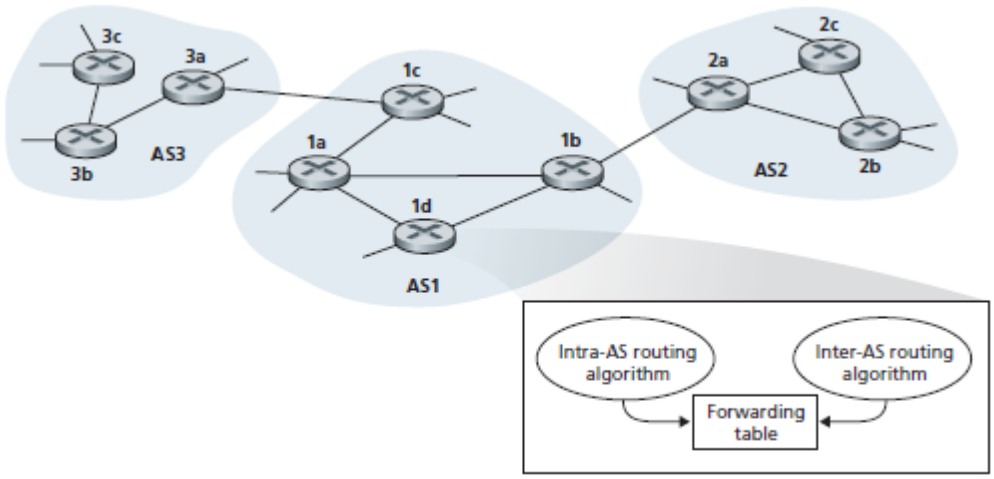


Figure 4.32 ♦ An example of interconnected autonomous systems

- **Intra-AS Routing** (Example – RIP, OSPF)

- **Inter-AS Routing** (Example – BGP – **iBGP** & **eBGP**)

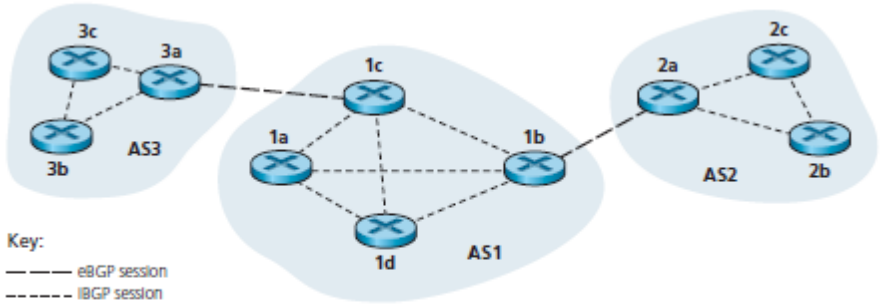


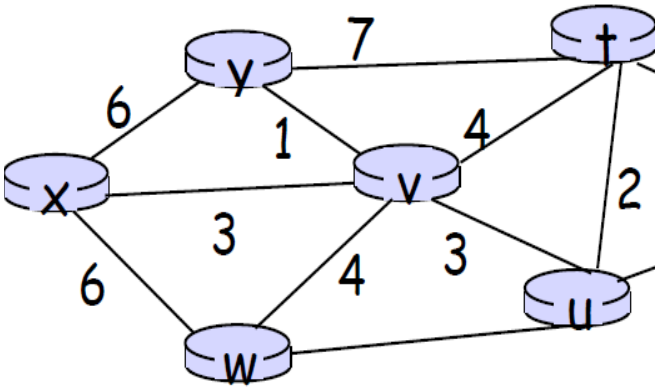
Figure 4.40 ♦ eBGP and iBGP sessions

Knowledge Test # 3

Q.1 Compare and contrast **link-state** and **distance-vector** routing algorithms **(Give 3 distinct points)**.

Link State algorithms	Distance-vector algorithms

Q.2 Consider the network shown below with link costs as indicated in the figure. To get credit for this question, you should **show all iterations** of the algorithm.



a) From the topology you have constructed, use **Dijkstra's algorithm** to compute the shortest path from **node Y** to all other nodes in the network (**Show all your calculations in every step**).

b) Construct the **routing/forwarding table** for **node Y**.

c) Now consider **Bellman-ford distance-vector algorithm**. Assume that each node only knows the costs to its direct neighbors initially (1st Step). Please provide **V's distance vector** to all other nodes in the network up to the **2nd step**. Please **show the steps of your computation.**

Q.3 What is an **Autonomous System (AS)**? Differentiate among **Intra-AS** and **Inter-AS** routing algorithms with **examples**.

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